

Velocity curve part 1 teacher notes

Galactic quadrants II and III

Help students find evidence of Galactic rotation using hydrogen spectra. Combine their measurements and interpret the Doppler-shift pattern.

Audience and prerequisites

Designed for advanced high school or introductory college students. Students should already read spectra, apply Doppler shifts, and understand relative velocity.

- Introduce the horn telescope and its spectrum display.
- Review electromagnetic waves and the 21 cm hydrogen transition.
- Practice celestial, Galactic, and horizontal coordinates.
- Review redshift, blueshift, and velocity along the line of sight.

Prepare the observation

Use one shared horn telescope or assign telescopes to groups. Set up before class and verify that the receiver and spectrum display work.

Pair students and assign one or two Galactic longitudes per pair. Cover $\ell = 90^\circ$ to 230° in 10° or 15° increments. Observe along the Galactic plane, where latitude $b = 0^\circ$.

Use Stellarium to choose a date and time when the assigned directions are visible from your site. Check the telescope's azimuth and elevation for each direction.

Teacher preview

[Watch Galaxy Survey Trailer](#)

The video scans Galactic longitudes from 90° to 240° . Its red reference line marks the unshifted hydrogen frequency near 1420.41 MHz. Compare the peak positions on either side of that line.

Measurement checks

Absolute intensity calibration is not required to compare peak frequencies. However, check frequency calibration, interference, and the spectral baseline. These affect whether a measured peak represents Galactic hydrogen.

Begin with a 10-second integration and increase it if the signal is unclear. Keep the settings consistent across the class survey.

Collect spectra and calculate velocities

1. Point the telescope toward the assigned Galactic coordinates.
2. Record the date, time, Galactic longitude, azimuth, and elevation.
3. Enter azimuth and elevation in the spectrometer fields when available. Confirm that saved filenames retain this information.
4. Use Capture Current Spectrum to save the measurement. Record the saved filename and integration time.
5. Repeat for the group's other assigned direction.

Measure the frequency shift

Display 1419.5 to 1420.5 MHz when this range contains the full signal. Use major gridlines every 0.1 MHz and minor gridlines every 0.01 MHz. Expand the range if a feature reaches its edge.

Identify the hydrogen peak and record its frequency. Estimate uncertainty from channel spacing, noise, and the peak's width. A 0.01 MHz grid interval corresponds to about 2.1 km/s; it does not establish measurement accuracy.

Convert frequency to radial velocity

$$v = c (f_0 - f_{peak}) / f_0$$

Use $f_0 = 1420.41$ MHz and $c = 3.00 \times 10^5$ km/s. Keep f_0 and f_{peak} in the same frequency units. This approximation applies when the measured speed is much smaller than c .

Result	Meaning
Positive velocity	Redshift: the gas is receding along the line of sight.
Negative velocity	Blueshift: the gas is approaching along the line of sight.

Have each group report longitude, quadrant, peak frequency, radial velocity, uncertainty, and redshift or blueshift. Record the velocity reference frame used by the software. Compare measurements only after checking that their reference frames agree.

Class results

Longitude	Quadrant	Velocity km/s	Uncertainty km/s	Redshift or blueshift

Interpret the class results

Look for a pattern

The simple circular-rotation model predicts blueshifted outer-Galaxy gas in quadrant II and redshifted gas in quadrant III. Quadrant II spans 90° to 180° ; quadrant III spans 180° to 270° . Near longitude 180° , the predicted rotational radial component approaches zero.

Earth's orbital motion can change uncorrected velocities by up to about 30 km/s. The projected correction depends on direction and observing date. Nearby gas, noncircular motions, and measurement uncertainty can also complicate the pattern.

Connect the pattern to Galactic rotation

For a nearly flat rotation curve, gas at larger radius can have a similar orbital speed to the Sun. Its angular speed is lower because $\omega = v/R$. The observed radial shifts are consistent with this difference in angular speed.

State the viewing direction when drawing clockwise or counterclockwise motion. Those words alone do not define a Galactic rotation convention.

Lead the discussion

1. Ask each group to present its measurements and uncertainty.
2. Combine the class results and compare the two Galactic quadrants.
3. Have groups explain their conclusions on whiteboards, using a labeled diagram.
4. Discuss Earth's motion, the velocity reference frame, and any peaks that differ from the simple model.
5. Ask what further observations would help test the interpretation.

Assess the explanation

Look for correct Doppler signs, units, coordinate use, and uncertainty. Students should distinguish radial velocity, orbital speed, and angular speed. Their conclusion should connect the class pattern to a stated model and acknowledge its limits.